

LANDSLIDE GEOHAZARD ASSESSMENT REPORT

Integrated Geodetic InSAR Diagnostics & Large Language Model Interpretability

Site Name:	TestRegion
Diagnostic ID:	sample_00002
Analysis Date:	2026-07-08 16:12:04
Geographic coordinates:	0.0°N, 0.0°E
Computed Risk Rating:	LOW

Prepared by the AI-Powered Geohazard Analytics System

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1. Executive Summary

An integrated geotechnical hazard assessment has been compiled for the surveyed site. Leveraging spaceborne multi-temporal interferometric synthetic aperture radar (InSAR) datasets and SegFormer-B2 computer vision models, our system has mapped active surface displacement boundaries. The computed core sliding zone is estimated at **0.0 m²**. Due to the steep geomorphic configuration and high precipitation triggers, the slope is classified at a **LOW** geohazard status, requesting prompt stabilization interventions.

2. Technical Analysis & Geodetic Metrics

The geodetic displacement vectors reveal downward line-of-sight (LOS) motion creep:

Interferometric Parameter	Monitored Value	Scientific Interpretation
Mean LOS Velocity	-49.88 mm/yr	Average downslope creep trend
Peak LOS Velocity	314.14 mm/yr	Critical maximum displacement velocity
Mean Acceleration	16.8 mm/yr²	Indicates accelerating/decelerating slope phase
Radar Coherence Index	0.5016990900039673	Correlation index measuring signal quality
Slope Angle / Aspect	0.0° / N/A	Gradient trigger values of monitored slide zone

3. Spatial Maps & Visual Results

The spatial distribution of active landslide boundaries compared across amplitude bands, prediction masks, heatmaps, and overlays:



Fig 1: Input InSAR Amplitude Map



Fig 2: SegFormer Prediction Mask



Fig 3: InSAR Displacement Heatmap



Fig 4: Blended Spatial Overlay

4. Emergency Recommendations & Mitigation Actions

- Immediate:** Install monitoring sensors in high-risk zones.
- Short-term:** Deploy drainage interventions to reduce pore pressure.
- Long-term:** Establish retaining structures and vegetation reinforcement.
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5. Plain-Language Summary

Satellite radar data shows ground movement on this slope. The AI model has identified active landslide areas covering approximately 0.0% of the monitored region. Risk level is LOW. Local authorities should monitor the area continuously.

6. Appendix & Geotechnical Metadata

This assessment is generated through the combination of interferometric satellite sensors and deep-learning report compilers.

Metadata Field	Value / Configuration	System Impact
SegFormer Backing	nvidia/segformer-b2-finetuned-ade	Active landslide body segmentation
Analysis Engine	Geodetic Extraction V1.4	Extracts area, coordinates, aspect, and slope stats
VLM Framework	Modular VLM Wrapper	Generates report text using image+text models
Failure Probability Score	0.2536708163352974	Computed using gradient & displacement ratio
Signal Coherence	0.5016990900039673	Measures phase quality of the radar backscatter

Prepared By:

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